Week 2 | **CLO** 2-C · 3-B · 3-C | **File** 4 / 7

2-3 Other Types of Graphs

رسوم أخرى — الأعمدة والأعمدة المركبة والدائرة والسلسلة الزمنية والساق والورقة

1 · Bar graph BOTH · CLO2-C

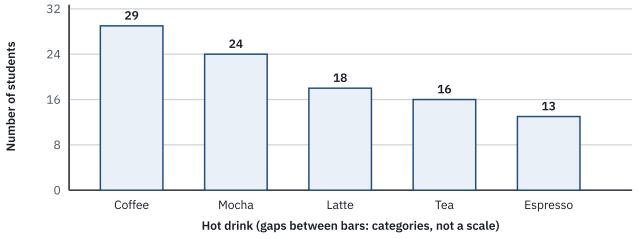
A **bar graph** represents data with **vertical or horizontal** bars whose heights or lengths are the frequencies (or relative frequencies, percentages, or the quantity measured). It is used for **categorical (qualitative)** data. Bars are equally spaced and **do not touch**, because the categories are separate rather than a continuous scale.

	HISTOGRAM	BAR GRAPH
Data	quantitative (grouped)	qualitative (categories)
x-axis	class boundaries, a number line	category names
Bars	touch	separated by gaps

HOT DRINKS, 100 STUDENTS AMANI

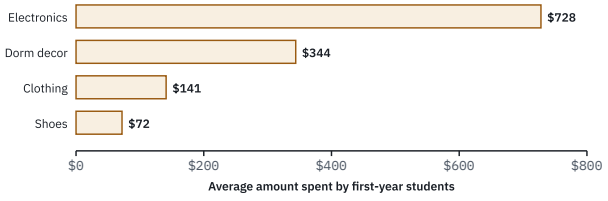
Make a frequency table and draw vertical and horizontal bar graphs.

DRINK	F
Latte	18
Espresso	13
Tea	16
Coffee	29
Mocha	24
Total	**100**



Dr. Amani's hot drinks example (n = 100), ordered here from most to least popular. Bars are separated because the x-axis holds categories, not a continuous scale. Coffee is the most preferred drink (29%), espresso the least (13%).

مثال المشروبات لـ د. أماني (n = 100). الأعمدة منفصلة لأن المحور الأفقي تصنيفات لا مقياس متصل. القهوة الأكثر تفضيلًا (29%) والإسبريسو الأقل (13%).

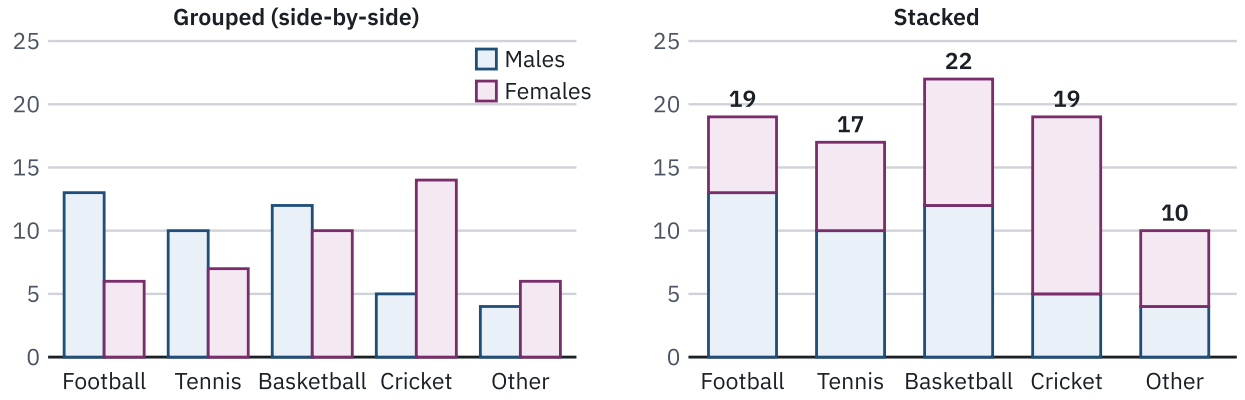


Example 2-8 as a horizontal bar graph: identical information to a vertical one, just rotated, and handy when category names are long. Electronics take more than the other three categories combined.

مثال 2-8 كأعمدة أفقية: نفس المعلومة لكن مستدارة، ومفيدة عندما تكون أسماء التصنيفات طويلة. الإلكترونيات أكثر من مجموع البنود الثلاثة الأخرى.

2 · Compound bar graphs BOTH

Bar graphs that compare data for **two or more groups**. A **grouped** (side-by-side) bar graph puts each group's bar next to the others; a **stacked** bar graph splits one column into sections, one per group.



Dr. Amani's compound bar graph data shown both ways. Grouped bars compare males with females inside each sport (cricket is the only sport where females clearly lead). Stacked bars show each sport's total (basketball 22) while keeping the split.

بيانات د. أماني بالشكلين: الأعمدة المتجاورة تقارن الذكور بالإناث داخل كل رياضة (الكريكيت الوحيدة التي تتفوق فيها الإناث بوضوح)، والأعمدة المكدسة تُظهر المجموع لكل رياضة (كرة السلة 22 مع الاحتفاظ بالتقسيم.

NEVER-MARRIED US ADULTS (MILLIONS) AHMED'S SLIDES

Males 15.3, 24.2, 32.3, 40.2 and females 12.3, 20.2, 27.8, 34.0 in 1960, 1980, 2000, 2010.

Both groups rise steadily every period, and males outnumber females every year. The stacked totals are 27.6, 44.4, 60.1 and 74.2 million, so the total nearly tripled from 1960 to 2010.

ACTIVITY · STACKED BAR GRAPH → FREQUENCY TABLE AMANI

Read the ggplot stacked bars (x = Child, Adult, Retired; groups A top, B middle, C bottom; tops at 48, 45, 29).

Each section's frequency = its **top minus its bottom**, not the top value.

CLASS	A	B	C	TOTAL
Child	48 − 35 = 13	35 − 16 = 19	16	48
Adult	45 − 34 = 11	34 − 15 = 19	15	45
Retired	29 − 21 = 8	21 − 14 = 7	14	29

3 · Pie graph BOTH

A circle divided into sections according to the **percentage** of frequencies in each category. Data are shown as percentages or proportions; the variable is

EXAMPLE 2-11 · SUPER BOWL SNACKS, N = 30.0 MILLION LB AHMED'S SLIDES

usually nominal or categorical; the purpose is to show the relationship of the **parts to the whole**.

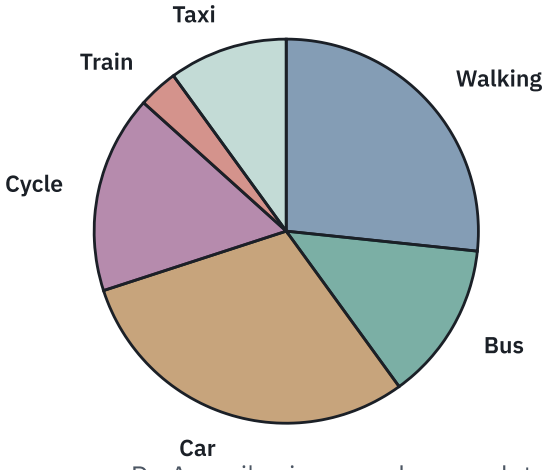
STEPS

$$\text{degrees} = \frac{f}{n} \times 360° \qquad \% = \frac{f}{n} \times 100$$

Draw each sector with a protractor and compass, and label it with the name and percentage.

SNACK	F	DEGREES	%
Potato chips	11.2	134.4°	37.3
Tortilla chips	8.2	98.4°	27.3
Pretzels	4.3	51.6°	14.3
Popcorn	3.8	45.6°	12.7
Snack nuts	2.5	30.0°	8.3
Total	**30.0**	**360°**	**99.9 ≈ 100**

The slide rounds 134.4° to 134°. Chips of both kinds make up almost two-thirds (64.6%).



Method	f	% (f/n·100)	degrees (f/n·360)
Walking	8	26.67	96°
Bus	4	13.33	48°
Car	9	30.00	108°
Cycle	5	16.67	60°
Train	1	3.33	12°
Taxi	3	10.00	36°
Total	**30**	**100**	**360°**

Dr. Amani's pie example, completed. Each angle is f/n × 360°, each percentage f/n × 100; the angles must add to 360° and the percentages to 100. Car (30%) and walking (26.7%) together cover more than half the class.

مثال الدائرة لـ د. أماني محلولًا: الزاوية = f/n × 360، والنسبة = f/n × 100، ومجموع الزوايا 360 ومجموع النسب 100. السيارة (30%) والمشي (26.7%) معًا أكثر من نصف الفصل.

4 · Time series graph BOTH

Represents data at **regular intervals over time**, as a line graph; useful for monitoring change. When analysing one, look for **trends** (upward or downward) and note the **slope** to see how fast the change is.

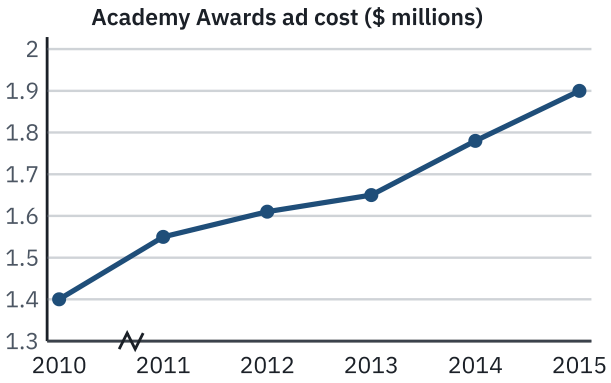
Steps: (1) time on the x-axis, the variable on the y-axis, with a suitable scale; (2) plot a point at each time; (3) join the points with straight lines to show the trend; (4) add a title and label both axes. A **compound time series** graph draws two or more

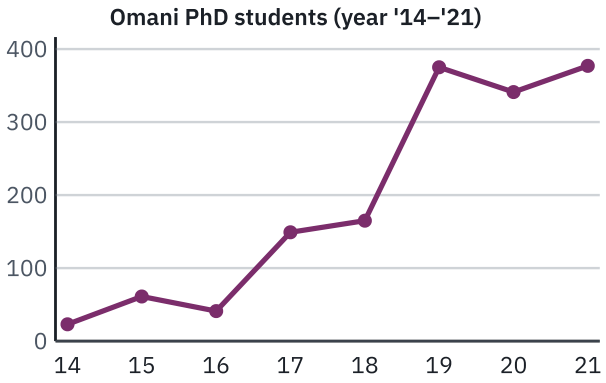
EXAMPLE 2-10 AND ACTIVITY BOTH

Academy Awards 30-s ad cost ($ m): 2010 1.40 · 2011 1.55 · 2012 1.61 · 2013 1.65 · 2014 1.78 · 2015 1.90. Omani PhD students (MHESRI): 2014 23 · 2015 61 · 2016 41 · 2017 149 · 2018 165 · 2019 375 · 2020 341 · 2021 377.

lines on one set of axes (e.g. the percentage of elderly men and women in the US labour force, 1960–2010). AHMED'S SLIDES

Ad cost: a steady **upward trend**, +$0.50 m over 5 years, rising every year. PhD students: strong overall growth from 23 to 377 (about 16 times), not steady: dips in 2016 and 2020, and the steepest rise from 2018 to 2019 (+210).





Left, Example 2-10: a steady upward trend, the cost rising every year (the y-axis starts at 1.3, so the zig-zag marks the break). Right, Dr. Amani's activity: overall strong growth, with dips in 2016 and 2020 and the steepest rise from 2018 to 2019 (+210 students).

يسار مثال 10-2: اتجاه صاعد ثابت كل سنة (المحور يبدأ من 1.3). يمين نشاط د. أماني: نمو قوي عمومًا مع انخفاض في 2016 و 2020، وأشد صعود بين 2018 و 2019 (+210 طالب).

5 · Stem-and-leaf plot BOTH

Shows **numerical** data arranged **in order**; for numeric data only. Each value is split into a **stem** (the leading digit or digits) and a **leaf** (usually the last digit), separated by "|". The plot needs a **key** to be readable (e.g. 2 | 5 means 25; 6 is recorded as 0 | 6). Its leaves act like horizontal histogram bars, so it shows the shape, and unlike a histogram it **keeps every original value**.

Steps: (1) arrange the data in ascending order; (2) choose the stems; (3) list the leaves beside each stem from smallest to largest, keeping a stem with no leaves; (4) give a key.

Decimals AMANI: 12.3, 12.5, 13.0 becomes 12 | 3 5 and 13 | 0 with key 12 | 3 = 12.3; 1.23, 1.25, 1.30 becomes 1.2 | 3 5 and 1.3 | 0 with key 1.2 | 3 = 1.23. A stem with no data still appears (her example has an empty "5 |" row).

EXAMPLE · 33 SCORES AMANI

100 65 78 83 79 92 95 67 71 85 83 · 92 86 63 91 86 75 78 73 86 91 77 · 79 80 91 87 85 76 68 76 89 96 98. (1) Shape? (2) Highest frequency (peak)?

```
 6 | 3 5 7 8
 7 | 1 3 5 6 6 7 8 8 9 9
 8 | 0 3 3 5 5 6 6 6 7 9
 9 | 1 1 1 2 2 5 6 8
10 | 0          key: 7 | 1 = 71
```

(1) Leaf counts 4, 10, 10, 8, 1: rises to a peak and falls, so **roughly bell-shaped (symmetric)**. (2) Peak **10** values, in both the 70s and the 80s.

ACTIVITY · DECIMALS AMANI

2.3, 2.5, 2.5, 2.7, 2.8, 3.2, 3.6, 3.6, 4.5, 5.0

```
2 | 3 5 5 7 8
3 | 2 6 6
4 | 5
5 | 0          key: 2 | 3 = 2.3
```

Shape: **right (positively) skewed**, with the counts 5, 3, 1, 1 falling away to the right. Peak **5** values, at stem 2.

EXAMPLE 2-14 · CARDIOGRAMS ON 20 DAYS AHMED'S SLIDES

25 31 20 32 13 14 43 02 57 23 36 32 33 32 44 32 52 44 51 45

```
0 | 2
1 | 3 4
2 | 0 3 5
3 | 1 2 2 2 2 3 6
4 | 3 4 4 5
5 | 1 2 7        key: 1 | 3 = 13
```

The peak is in the 30s (7 days) with a longer tail toward low values (2, 13, 14): **slightly left-skewed**, as the lecturer marked. The first plot on the slide has unordered leaves; always reorder them.

EXAMPLE 2-15 · SPLIT STEMS: CAR THEFTS ON 30 DAYS AHMED'S SLIDES

52 62 51 50 69 58 77 66 53 57 75 56 55 67 73 79 59 68 65 72 57 51 63 69 75 65 53 78 66 55

```
5 | 0 1 1 2 3 3        (leaves 0–4)
5 | 5 5 6 7 7 8 9      (leaves 5–9)
6 | 2 3
6 | 5 5 6 6 7 8 9 9
7 | 2 3
7 | 5 5 7 8 9          key: 5 | 2 = 52
```

With only three stems (5, 6, 7) the ordinary plot is too coarse: 13, 10 and 7 leaves. **Splitting** each stem into a low half (leaves 0–4) and a high half (leaves 5–9) shows more detail: most days had 55–69 thefts, and there are noticeably few in the low 60s (62–63) and low 70s.

الشرح بالعربي — الرسوم الأخرى

(1) **الأعمدة** للبيانات الوصفية، وأعمدتها **منفصلة** (بخلاف المدرج التكراري الذي أعمدته متلاصقة). (2) **الأعمدة المركبة**: متجاورة للمقارنة داخل كل فئة، أو مكدسة لإظهار المجموع. وعند قراءة المكدسة: تكرار كل جزء = **أعلاه − أسفله**، لا قيمة أعلاه. (3) **الدائرة**: الزاوية = `f/n × 360` والنسبة = `f/n × 100` ، ومجموع الزوايا 360. (4) **السلسلة الزمنية**: الزمن على المحور الأفقي، ونصف **الاتجاه** (صاعد/هابط) و**سرعة التغير** (الميل). (5) **الساق والورقة**: نرتب البيانات، الساق للخانات الأولى والورقة للخانة الأخيرة، ولا بد من **مفتاح (key)**، ولا نحذف ساقًا فارغة. ميزته أنه يحتفظ بكل القيم الأصلية ويُظهر شكل التوزيع.

Chapter 2 study set: 7 files. One merged source built from Dr. Amani Al-Rubkhi's Fall 2026 notes (course coordinator) and Dr. Ahmed Al-Hadhrami's slide pack, aligned to the SQU STAT 2101 course outline (Fall 2026). Textbook: Bluman, *Elementary Statistics*, 10th ed.

00 · Exam brief · 01 · Lesson 2-1: Organizing Data · 02 · Lesson 2-2: Histograms, Frequency Polygons, Ogives · **03 · Lesson 2-3: Other Types of Graphs** · 04 · Revision pack · 05 · Exam-style questions · 06 · Full worked answers